

Plantim Icons — v3

Process report: building an additive multi-color, multi-size icon set

Branch **feat/icons-v3-multicolor-sizes** · commit **6ee40d8** · v2 registry **v2.0.1** · v3 **v3.0.0** · v3Hash 61fc82579babf4b1...

Scope: **237 icons** × (4 styles + 3 multicolor sizes 24/48/72px) = **1659 SVG files** · stroke policy **proportional**.

Why this exists

The package centralizes icons for Plantim's web + mobile stack ahead of the Android MVP. The existing v2 set (154 icons) was generated by Codex and is not fully trusted. Rather than mutate the heavily-governed v2 registry, v3 is built **alongside** it: a trusted, human-inspectable, portfolio-ready set that adds two things v2 cannot express — **three baked sizes per icon** and **one multi-color SVG per icon**.

Decisions

Decision	Choice	Rationale
v3 vs v2	Additive alongside v2	v2 registry, exports, and CI stay untouched and green; safest to centralize on and to showcase.
Sizes	24 / 48 / 72 (1x/2x/3x)	Matches the "3x bigger" ask; exposed as a <code>sizes</code> property and as separate files.
Multi-color	Category palette	Systematic and scalable across 154 icons; consistent per family, with a small semantic override table.
Reports	2 PDFs	This process report + a visual showcase catalog.
Geometry	Derived from v2	v2 <code>nodes []</code> are the single geometry source, so v3 can never drift — except explicit, documented corrections (below).

How a monochrome icon becomes multi-color

Every v2 icon is stroke-only (`stroke="currentColor"`). The generator assigns each node a **role**, then a **color** from the icon's category palette (first match wins):

- **accent** — a point/dot: a `circle` with $r \leq 2$, a near-zero-length `line` (how Lucide draws alert dots, e.g. `x1=12 x2=12.01`), or a tiny dot path.
- **secondary** — a container/frame: `rect`, `ellipse`, or `circle` with $r \geq 8$ (outer rings).
- **primary** — everything else (the main glyph).

Refinements: an icon with no primary node (e.g. an all-rectangle grid) has its secondaries promoted so it is never fully muted; `plant` icons use **stem two-tone** (first stroke = stem/primary, rest = foliage/secondary); tinted categories add a faint bg-filled shape behind containers; `status` icons keep **semantic color** (success green, warning amber, error red, info blue).

Worked example — `status.error`



v2 monochrome



v3 multicolor

Outer ring → secondary + neutral tint disc · vertical stroke → primary (semantic red) · indicator dot → accent.

Four styles + animation

Every icon renders in four styles from the same geometry: **outlined** (currentColor line), **solid** (filled glyphs — frames/rings stay open so inner detail is never hidden), **colored** (single category hue), and **multicolor** (per-role palette). A reusable stylesheet (`animations.css`) adds spin / pulse / beat / bounce / wiggle / grow / draw animations usable on any icon. All four styles, sizes, and animations are browsable and searchable in `reports/plantim-icons-v3-catalog.html`.

New icons for Plantim (83)

Grounded in the Plantim app codebase, v3 adds **83 brand-new icons** on top of the 154 derived from v2. Highlights: the **Today** tab no longer has to borrow the Home icon (three alternatives provided); **fertilize** and **repot** — which both reused the leaf in v2 — get dedicated icons; care types match the app's authoritative set (water, fertilize, prune, repot); and each of the **16 plant families** enumerated in the app has its own botanical icon. Tab concepts each ship multiple alternatives so a final icon can be chosen by comparison (see the showcase catalog).

Group	Icons
Today	3
Garden (tab)	3
Plant (tab)	3
Calendar (tab)	3
Feed (tab)	3
Chat (tab)	3
Care actions	11
Plant lifecycle	3
Plant health	3
Growth stages	3
Garden tools	14
Plant families	16
Taxonomy & genus	15

All new icons flow through the same multicolor + 24/48/72 pipeline and are browsable/searchable in `reports/plantim-icons-v3-catalog.html`.

Findings — corrected v2 geometry

Reproducing v2 geometry surfaced a genuine defect, validating the distrust of the Codex-generated set. **1 icon(s)** had clearly-broken geometry, corrected in v3 only (v2 registry untouched; each correction is declared in `palette.json` → `geometry0verrides` and is reversible):

weather.sun

fixed



v2 (broken)

v3 (corrected + colored)

v2 stored all 8 sun rays as zero-length dots (`d="M12 4h.01"`), so the sun rendered as a dotted ring with no rays. Restored to correct Lucide ray strokes.

Note: three other icons (`utility.calendar.days`, `utility.cookie`, `utility.cloud.snow`) also use zero-length "dot" paths, but those are **legitimate** Lucide features (day markers, cookie chips, snow) and were left as-is.

Category palette

Category	Primary	Secondary	Accent	Bg tint	Icons	Style
plant	 #2E7D32	 #66BB6A	 #FBC02D	 #E8F5E9	8	stem
weather	 #0288D1	 #4FC3F7	 #FFB300	 #E1F5FE	3	tint
status	 #455A64	 #90A4AE	 #D32F2F	 #ECEFF1	9	tint
navigation	 #37474F	 #78909C	 #1E88E5	 #ECEFF1	8	—
action	 #37474F	 #78909C	 #00897B	 #ECEFF1	4	—
account	 #5E35B1	 #9575CD	 #00897B	 #EDE7F6	2	tint
calendar	 #3949AB	 #7986CB	 #E53935	 #E8EAF6	2	tint
location	 #C2185B	 #F06292	 #FBC02D	 #FCE4EC	1	tint
utility	 #455A64	 #90A4AE	 #1E88E5	 #ECEFF1	117	—
nav	 #2E7D32	 #66BB6A	 #FB8C00	 #E8F5E9	18	tint
garden	 #558B2F	 #8D6E63	 #FBC02D	 #F1F8E9	14	tint
care	 #00897B	 #4DB6AC	 #FB8C00	 #E0F2F1	11	tint
lifecycle	 #2E7D32	 #A5D6A7	 #FB8C00	 #E8F5E9	6	stem
health	 #455A64	 #90A4AE	 #D32F2F	 #ECEFF1	3	tint
family	 #33691E	 #7CB342	 #FBC02D	 #F1F8E9	16	stem
genus	 #1B5E20	 #66BB6A	 #FBC02D	 #E8F5E9	15	stem

A small override table also gives semantically-colored icons their natural hues (e.g. `weather.sun` → amber, `utility.flame` → orange, `utility.star` → gold, `utility.heart` → red).

Validation results

v3 self-check pass

```
$ node design-tokens/bin/check-v3-icons.mjs
v3 self-check passed: 237 icons, 1659 SVG files (4 styles + 3 sizes), hash OK.
```

v2 left untouched verified

No tracked change to the v2 registry or any generated adapter path. New files are entirely additive.

```
$ git diff --stat -- design-tokens/icons/registry.json packages/plantim-icons/src packages/plantim-
icons/dist packages/PlantimIcons/Sources
(no changes – v2 adapter paths are untouched)
```

```
Modified tracked files touching v2 paths: 0
New (untracked) top-level areas: design-tokens/icons, reports/PR-description.md
```

Full v2 suite

```
$ npm run icons:validate:node
Generated 154 Plantim icons from Lucide geometry.
Generated native Swift renderer for 154 icons.
Generated MIGRATIONS.md with 117 compatibility aliases.
Icon contract valid: 154 icons.
Icon migration contract valid: 117 aliases.
Icon registry tests passed: 154 icons.
Status icon labels exist in every locale: 6 keys.
Generated web and iOS icon galleries are wired to the canonical registry.
Vue API behavior valid for 154 icons.
Package compatibility contract valid: runtime Node >=18, Vue >=3; tooling Node >=22.18/npm >=11.12.
Release governance valid: provenance, pinned actions, public access, and documented trusted-publisher
transition.
npm package is publish-ready: @plantim/icons@2.0.1 (11 files).
Icon package API valid: @plantim/icons@2.0.1.
Icon package size valid: entry 198310 bytes (18951 bytes gzip), dist 225308 bytes.
```

Governance safety

Every existing v2 check is path-scoped; none globs the tree for SVGs. v3 stays green with zero edits to any v2 validator or schema because it never:

- touches `registry.json` or its hash (the v2 generator re-verifies it);
- adds itself to `packages/plantim-icons/package.json` → `files[]/exports`, so the npm tarball, the ≤30 KiB gzip size budget, and the public API check are unaffected;
- writes into the CI diff-checked adapter paths (`packages/plantim-icons/{src,dist}`, `packages/PlantimIcons/Sources`);
- alters the `icons:validate:node` chain — v3 is exposed only through new, opt-in scripts.

What was produced

```
design-tokens/icons/v3/
  palette.json          hand-authored config (palette, sizes, roles, overrides)
  authored.mjs          38 hand-authored new icons (tabs, care, lifecycle, health, stages)
  custom-src/           45 new SVG-authored icons (garden tools, families, genus) + icons.json
  animations.css        reusable animations (spin/pulse/beat/bounce/wiggle/grow/draw)
  registry.v3.json, index.v3.json  generated superset + manifest
  {outlined,solid,colored,multicolor}/*.svg  237 masters each (4 styles)
  sizes/*@{24,48,72}.svg  711 multicolor size files
design-tokens/bin/
  generate-v3-icons.mjs, lib/v3-transform.mjs, check-v3-icons.mjs
  generate-v3-report.mjs, generate-v3-catalog.mjs, generate-v3-html.mjs, lib/html-to-pdf.mjs
reports/
  plantim-icons-v3-report.pdf, plantim-icons-v3-catalog.pdf
  plantim-icons-v3-catalog.html  (all icons × 4 styles, searchable + animated)
```

Next steps

- Review the companion **showcase catalog** PDF for the full visual set.
- If v3 graduates to a shipped format, add a dedicated export/entry (kept out of the v2 package) and a v3 gallery + drift check in CI.
- Consider a broader geometry audit of v2 for other Codex artifacts before the Android MVP consumes these.